Zack Valavanis

Chicago, IL | 847-730-7612 | zval321@gmail.com | LinkedIn | GitHub | Portfolio

Software engineer with a finance background transitioning into full-stack development. Experienced building REST APIs with FastAPI and Python, React/TypeScript frontends, and PostgreSQL-backed systems. Brings a unique edge in data-heavy and fintech domains. Pursuing a BS in CS (CU Boulder, 4.0 GPA).

PROJECTS

StatDiamond — Baseball Analytics Platform | *FastAPI · PostgreSQL · SQLAlchemy · React · TypeScript · Supabase · JWT Auth · pybaseball*

- Built a full-stack baseball stats app with FastAPI backend and React/TypeScript frontend, serving live standings and player/team data via pybaseball.
- Implemented JWT auth with token-based login, protected routes, and role-scoped access control using SQLAlchemy models and Pydantic schemas.
- Designed normalized PostgreSQL schema on Supabase with relational models for teams, players, and users; exposed full CRUD REST API.
- Built dynamic React UI with team/player filtering and standings tables — deployed on Vercel.

AlgoLedger — Algorand Portfolio Analytics & Tax Tool | *FastAPI · React · Claude Code · TypeScript · Supabase · PostgreSQL · Vercel · Railway*

- Built and deployed a freemium Algorand portfolio analytics dashboard with FIFO/LIFO/HIFO tax lot tracking, serving 200+ users with 6 registered accounts since launch.
- Engineered FastAPI backend with Supabase PostgreSQL, implementing Row-Level Security (RLS) policies for multi-tenant data isolation and secure API access.
- Designed responsive React/TypeScript frontend with real-time portfolio valuation, transaction history, and exportable tax reports for crypto investors.
- Deployed full CI/CD pipeline with Railway (backend) and Vercel (frontend); promoted via Reddit and X to drive organic user acquisition.

EXPERIENCE

Data Analytics Analyst / Client Associate 2023 – Present

Neuberger Berman · Chicago, IL

- Developed automation scripts (VBA/Python) to streamline financial reporting, reducing manual workload by 56 hours annually.
- Managed a $2.7B diversified portfolio, delivering customized client presentations tracking KPIs such as risk-adjusted returns, sector allocations, volatility, and benchmark performance.
- Created detailed, data-driven client reports and dashboards tailored to portfolio goals; applied VBA-based validation tools to ensure reporting accuracy and integrity.
- Provided outstanding customer-facing support by addressing client needs, offering tailored solutions, and building strong, lasting relationships.

Real Estate Analyst 2022 – 2023

Accenture · Chicago, IL

- Managed a $1.1B portfolio of 37 commercial office spaces, improving tenant satisfaction by 20% and driving operational efficiencies across multiple sites.
- Streamlined vendor operations and implemented process improvements, achieving $200K in savings and a 5% reduction in operating expenses.
- Managed the transition of 12 properties, collaborating with the real estate management team to optimize vendor management and rent processes.

Risk Analyst 2020 – 2022

Advantage Futures · Chicago, IL

- Streamlined quarterly reporting by implementing VBA macros to automate data compilation, reducing reporting time by over 50%.
- Managed analysis of volume and open interest in emerging exchange-traded products including Bitcoin and Ethereum futures, improving portfolio performance tracking by 15%.
- Conducted data analysis on margin usage, reducing risk exposure by 20% and identifying optimization opportunities for capital allocation.

OTHER EXPERIENCE

Founder & Frontend Engineer Summer 2025

Salus Wellness — AI-powered personal training & workout application (React Native)

- Designed and built core app screens in React Native from Figma prototypes, using AI-assisted code generation to accelerate development.
- Owned frontend implementation and debugging across the app's UI and state management.
- Collaborated within a founding team of 7-8 to build toward launch before the project wound down due to funding and engineering resource constraints.

Software Engineering Intern — Frontend Summer 2025

Capture — Privacy-first social media application (React Native)

- Built and refined React Native screens and UI components as part of a 3-person early-stage engineering team.
- Gained hands-on experience with React Native state management and mobile UI implementation in a fast-moving startup environment.

EDUCATION

University of Colorado Boulder — Post-Baccalaureate in Computer Science (In Progress) · GPA: 4.0

Actualize Coding Bootcamp — Full-Stack Web Development: Ruby on Rails, JavaScript, React

University of Colorado Boulder — B.S. Business Administration, Leeds School of Business · Chancellor's Achievement Scholar

University of Westminster, London — Study Abroad, Junior Year

SKILLS & CERTIFICATIONS

Languages: Python, JavaScript, TypeScript, C++, Ruby, SQL, VBA

Frameworks: FastAPI, React, React Native, Ruby on Rails, Node.js, Express

Databases & Tools: PostgreSQL, SQLAlchemy, Supabase, Git, REST APIs, Vercel, Heroku, Jest

Certifications: FINRA Series 3, 7, 63, 65 · Google Project Management · Tableau · Blockchain & Digital Assets (CBDA)